

CONSTANCE GOSSELIN

Geospatial Data Scientist | Climate & Nature Risk Modelling | Earth Observation

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PROFILE

Geospatial Data Scientist specialized in translating multi-source Earth Observation and environmental datasets into operational risk intelligence. Expert in designing reproducible spatial ETL pipelines, spatial statistics, and machine learning models for climate risk adaptation, hazard mapping, and physical risk assessment.

CORE EXPERTISE

- Climate Risk Modelling
- Nature-Risk Assessment
- Hazard & Exposure Mapping
- Spatio-Temporal Analysis
- Earth Observation (EO)
- Spatial Machine Learning
- Reproducible Pipelines
- Terrain & Surface Analysis

TECHNICAL SKILLS

- Programming
- Python R Git Linux
- EO & Cloud Computing
- Google Earth Engine
- Copernicus API Sentinel-1 (SAR)
- Sentinel-2 Landsat
- ERA5 Reanalysis
- Spatial Data Stack
- GeoPandas Xarray Rasterio
- Shapely QGIS ArcGIS Pro

LANGUAGES

- French (Native) ● ● ● ● ●
- English (Fluent - C1) ● ● ● ● ●
- Spanish (Working - B1) ● ● ● ● ●

PROFESSIONAL EXPERIENCE

Geospatial Data Scientist – Climate Risk Modelling

University of La Laguna

- 📅 03/2026 – Present Tenerife, Spain
- Develop automated end-to-end Python and R pipelines integrating high-resolution Copernicus products and climate reanalysis data.
 - Engineer multi-hazard spatial indicators (swell directions and intensity, pressure, wind directions, coastal vulnerability metrics) to support regional resilience frameworks.
 - Apply spatial machine learning models to environmental classification and predictive physical hazard assessments.

Remote Sensing & Geospatial Data Scientist

UiT The Arctic University of Norway

- 📅 02/2024 – 07/2024 Tromsø, Norway
- Built automated EO analytics workflows analyzing multi-decadal time-series data from Sentinel and Landsat constellations.
 - Conducted advanced statistical modeling (SARIMA, spatio-temporal trends) to track climate-driven ocean and coastal dynamics.
 - Developed high-impact, decision-ready geospatial layers to guide maritime and coastal resilience.

Geospatial Data Engineer – Climate Analytics

University of Namur

- 📅 06/2023 – 08/2023 Namur, Belgium
- Built high-performance spatial ETL pipelines integrating heterogeneous datasets (NASA climate grids, gridded demographic layers).
 - Modeled socio-environmental exposure risks using predictive statistical frameworks for hazard-driven events.

SELECTED GEOSPATIAL PROJECTS

Natural Hazard & Human Mobility Framework

Causal Inference & Large-scale Spatial Analysis

- Developed a robust quasi-experimental framework using the 2023 Türkiye earthquake as a hazard shock to model human response.
- Integrated Meta high-frequency mobility metrics with remote sensing damage indicators and socio-economic vulnerability factors.
- Applied spatial causal inference and sensitivity analyses to output resilient disaster response patterns.

Wildfire Susceptibility Modelling

Dynamic Risk Mapping

- Built a predictive wildfire susceptibility model using GEE, integrating MODIS fire anomalies, NDVI dynamics, and meteorological metrics.
- Created dynamic risk evaluation maps directly interpretable for active forest management and local safety actors.

RESEARCH OUTPUT

Working Paper (Under Review):
"Meta user movement data for analysing disaster-related mobility: Sensitivity, bias and usability." Submitted to **Spatial Information Research**.

INTERNATIONAL EXPERIENCE

Erasmus Course
Gothenburg University
📅 2021-2022

High school exchange
James Hargest College
📅 2015-2016

Urban Heat Island & Microclimate Analysis

Cloud-native Earth Observation

- Engineered automated Google Earth Engine (GEE) workflows utilizing Landsat 5/8/9 thermal bands.
- Modeled Land Surface Temperature (LST) and normalized vegetation structures to map urban vulnerability hotspots.
- Validated satellite-derived thermal indicators using heterogeneous in-situ sensor telemetry.

EDUCATION

M.Sc. in Geography (Global Change focus)
University of Liège
📅 09/2024 – 01/2025 Belgium
Graduated with Distinction. Advanced coursework in spatial statistics, environmental modelling, GIS, remote sensing and climate change adaptation.

B.Sc. in Geography
University of Namur
📅 09/2018 – 09/2022 Belgium